

Velkscale Modeling Report - 25 September 2026



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- **Bold issues** indicate those needing attention or currently in the process of being fixed. Warnings are especially notable.
- Treat current proxy models as a heuristic.

Currently implemented code creates a sequence of estimators, prior-based transformations, and accounting adjustments in ensemble form a jointly solved demographic system.

Its central accounting equation is:

$$\frac{\Delta P_{i,t}}{h_t} = B_{i,t} - D_{i,t}^f - D_{i,t}^m + M_{i,t},$$

where (i) is a raster pixel, (t) is a year, and (h_t) is the interval represented by the population delta.

The implementation does not guarantee that this identity and global zero-sum migration both hold absolutely in final saved outputs, only that it held approximately (in testing) due to issues with convergence and floating-points. This was accepted for the sake of performance and compute time.

1. Notation and dependency order

Let:

- $(P_{i,t})$: Stadestér population.
- $(\Delta P_{i,t})$: supplied population-change raster.
- $(N_{i,t,a}^s)$: population of sex (s) in age band (a).
- $(B_{i,t})$: annual births.
- $(D_{i,t}^s)$: annual deaths by sex.
- $(M_{i,t}^s)$: annual net migration by sex.
- $(M_{i,t} = M_{i,t}^f + M_{i,t}^m)$.
- $(X_{i,t})$: land-use, population, economic, and delta covariates.

The age bands are:

$$a \in \{00, 01, 05, 10, \dots, 80\},$$

with coded widths and midpoints:

$$w_a = (1, 4, 5, \dots, 5),$$
$$x_a = (0.5, 3, 7.5, 12.5, \dots, 82.5).$$

Warning

For technical reasons, the last band is treated computationally as five years wide, even though the source cohort represents 80+, since it has an unbounded upper domain and the surviving cohort is relatively small.

The dependency order implied by the files is:

1. Construct age–sex populations.
2. Construct an initial migration estimate.
3. Construct births and deaths, then reconcile them against that migration estimate.

The shared orchestrator for this order is run out of the `NWM` class.

2. Age–sex populations

2.1 Select age–sex training targets

The training target for cohort $((s,a))$ is selected by year:

$$Y_{i,t}^{s,a} = \begin{cases} Y_{i,t}^{s,a,\text{HMD}}, & t < 1950, \\ Y_{i,t}^{s,a,\text{UNWPP}}, & 1950 \leq t < 2015, \\ Y_{i,t}^{s,a,\text{WorldPop}}, & t \geq 2015, \end{cases}$$

with UNWPP fallback if the WorldPop file is unavailable.

The default training domain is 1750–2025.

2.2 Estimate cohort probabilities

The code delegates training to `trainMultinomialLogitModelsParallel`. The intended multinomial-logit prediction is:

$$p_{i,t|u}^{s,a} = \frac{\exp\left(b_u^{s,a} + X_{i,\tau(t)}^\top \beta_u^{s,a}\right)}{\sum_{r,c} \exp\left(b_u^{r,c} + X_{i,\tau(t)}^\top \beta_u^{r,c}\right)},$$

where (u) is the model's training year and:

$$\tau(t) = \min(t, 2023).$$

The default configuration disables the intercept and sets regularisation to zero. Softmax is the intended model equation for multinomial logit, but there are different variations of this. Currently, softmax is the version used by Velkscale.

Also, despite the comment claiming full-raster training, the supplied default is `sample_limit: 15000`.

2.3 Weight temporal models

For a model trained at year (u) , let (n_u) be its reported training sample count. Its coverage score is:

$$c_u = \left(\frac{n_u}{\max_v n_v} \right)^{0.3},$$

and its unified ensemble weight is:

$$q_u = \frac{c_u}{\sum_v c_v}.$$

For prediction year $(t \geq 1500)$, apply an **exponential**, not Gaussian, temporal kernel:

$$k_{t,u} = \exp \left(-\frac{|t - u|}{50} \right),$$

$$\omega_{t,u} = \frac{q_u k_{t,u}}{\sum_v q_v k_{t,v}}.$$

If a model exists for the prediction year itself, the final weights become:

$$\tilde{\omega}_{t,u} = 0.8 \omega_{t,u} + 0.2 \mathbf{1}_{u=t}.$$

Otherwise:

$$\tilde{\omega}_{t,u} = \omega_{t,u}.$$

If the raster helper averages model probabilities, the resulting probability is:

$$p_{i,t}^{s,a} = \sum_u \tilde{\omega}_{t,u} p_{i,t|u}^{s,a}.$$

That last aggregation rule needs confirmation from the helper implementation. It should be specified that the 80/20 mix is exapted from the Pareto principle, i.e. a generalisation of the Pareto distribution. In earlier tests, this was found to give the best R^2 fit.

For ($t < 1500$), the code selects the external premodern model if present. Its construction is not shown.

2.4 Collapse sex-specific predictions into age totals

For each pixel, sanitize probabilities and sum them by age:

$$T_a = [p^{m,a}]_+ + [p^{f,a}]_+,$$

where here $([z]_+)$ means positive finite values are retained; other values become zero.

Let (r_a) be the fixed male/female ratio:

$$r_a = (1.045, 1.042, 1.040, 1.038, 1.030, 1.015, 0.985, 0.975, 0.965, 0.960, 0.950, 0.930, 0.900, 0.860, 0.810, 0.770, 0.740, 0.710, 0.680, 0.650, 0.620, 0.590, 0.560, 0.530, 0.500, 0.470, 0.440, 0.410, 0.380, 0.350, 0.320, 0.290, 0.260, 0.230, 0.200, 0.170, 0.140, 0.110, 0.080, 0.050, 0.020, 0.000)$$

Warning

Such rates are derived from weighted Coale-Demeny and HMD lifetables. Unfortunately, this is an anachronistic abstraction prior to 1950 for older-age cohorts, and should be replaced in future iterations of Velkscale.

Define:

$$W_a = w_a(1 + r_a),$$

$$z_a = \frac{T_a}{W_a}.$$

Thus (z_a) is the **female density implied by the imposed ratio**, not necessarily the female density originally predicted by the model.

2.5 Apply weighted isotonic regression

The caller uses `start_index = 2`, so the first two age bands remain outside the monotonicity constraint. This is used to reflect the bathtub curve (i.e. in three-component Siler mortality models).

For bands from age 5 onward:

$$\hat{\mathbf{z}} = \underset{u_2 \geq u_3 \geq \dots \geq u_{17}}{\operatorname{argmin}} \sum_{a=2}^{17} W_a (u_a - z_a)^2,$$

with:

$$\hat{z}_0 = z_0, \quad \hat{z}_1 = z_1.$$

There is no constraint connecting band 1 to band 2.

Reconstitute sex-specific amounts:

$$F_a^* = w_a \hat{z}_a,$$
$$M_a^* = r_a w_a \hat{z}_a.$$

Consequently:

$$\frac{M_a^*}{F_a^*} = r_a$$

where the denominator is positive.

This fixes the sex ratio exactly; it does not merely bound it within an envelope.

2.6 Scale to the population aggregate

Let:

$$Z = \sum_a (F_a^* + M_a^*).$$

For ($Z > 0$):

$$\hat{N}_{i,t,a}^f = P_{i,\tau(t)} \frac{F_a^*}{Z},$$
$$\hat{N}_{i,t,a}^m = P_{i,\tau(t)} \frac{M_a^*}{Z}.$$

If ($Z = 0$), the fallback is:

$$\hat{N}_{i,t,a}^s = \frac{P_{i,\tau(t)}}{36}.$$

Pixels with population below (0.01), or NaN population, are left zero.

Before rounding, for processed pixels:

$$\sum_{s,a} \hat{N}_{i,t,a}^s = P_{i,\tau(t)}.$$

Notice that **post-2023 modelled age structures are scaled to 2023 population**, not necessarily the requested year's population.

2.7 Composite the final population pyramid

Finally:

$$N_{i,t,a}^s = \begin{cases} Y_{i,t}^{s,a,\text{UNWPP}}, & 1950 \leq t < 2015, \text{ file available,} \\ Y_{i,t}^{s,a,\text{WorldPop}}, & 2015 \leq t \leq 2025, \text{ file available,} \\ Y_{i,t}^{s,a,\text{UNWPP}}, & 2015 \leq t \leq 2025, \text{ fallback available,} \\ \hat{N}_{i,t,a}^s, & \text{otherwise.} \end{cases}$$

These source files are copied **after** the modelled-cohort clamping stage. Their consistency with Stadestér is therefore inherited from upstream processing, not enforced again here.

3. Construct demographic denominators for births and deaths

The vital-statistics models do not directly predict event counts. They predict multipliers of structural denominators, i.e. what the delta population plainly attributable to a vital statistic stock (births, deaths, net migration).

3.1 Birth denominator: infant population

$$H_{i,t}^B = N_{i,t,00}^f + N_{i,t,00}^m.$$

The model will reconstruct births as:

$$B_{i,t} \approx \theta_{i,t}^B H_{i,t}^B.$$

The multiplier is capped at 1.5.

This is a prior relationship between infant stock and annual births—not an accounting identity.

3.2 Death denominator: growth-adjusted age-profile decline

For each sex, first divide each cohort population by its coded age-band width:

$$A_{i,t,a}^s = \frac{N_{i,t,a}^s}{w_a}.$$

Estimate total-population growth using the previous year in the full HYDE sequence (10000BC-2025AD):

$$g_{i,t} = \frac{\log(P_{i,t}/P_{i,t^-})}{t - t^-}.$$

The implementation substitutes zero growth when the previous population is nonpositive.

Construct a growth-adjusted age profile:

$$S_{i,t,a}^s = A_{i,t,a}^s \exp(g_{i,t} x_a).$$

Then apply decreasing isotonic regression across all age bands:

$$\hat{\mathbf{S}}^s = \underset{u_0 \geq u_1 \geq \dots \geq u_{17}}{\operatorname{argmin}} \sum_a w_a (u_a - S_a^s)^2.$$

For every nonterminal band:

$$\mu_a^s = \begin{cases} \frac{\hat{S}_a^s - \hat{S}_{a+1}^s}{\hat{S}_a^s (x_{a+1} - x_a)}, & \hat{S}_a^s > 0, \\ 0, & \text{otherwise.} \end{cases}$$

For the terminal band:

$$\mu_{17}^s = \begin{cases} 1/5, & \hat{S}_{17}^s > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The death denominator is:

$$H_{i,t}^{D^s} = \sum_a N_{i,t,a}^s \mu_{i,t,a}^s.$$

The model then reconstructs deaths as:

$$D_{i,t}^s \approx \theta_{i,t}^{D^s} H_{i,t}^{D^s}.$$

Warning

This is a mortality proxy under strong assumptions. A cross-sectional age bulge need not be migration; it can reflect past fertility or mortality changes. PAVA cannot uniquely remove migration or recover a true survival curve. Likewise, recent local population growth need not equal the stable-population growth rate required by the survival interpretation.

We have tried previously to use intercepts, this resulted in over-homogenisation. **It is therefore the recommendation of Demography to move this to a T-LSTM Neural Network model**, since it deals with this exact type of fitting problem.

4. Training targets for vital events and migration

4.1 Birth and death multiplier targets

For $(v \in \{B, D^f, D^m\})$, select source event counts:

$$V_{i,t}^{v,\text{src}} = \begin{cases} V_{i,t}^{v,\text{HMD}}, & t < 1950, \\ V_{i,t}^{v,\text{UNWPP}}, & t \geq 1950. \end{cases}$$

Then:

$$y_{i,t}^v = \min \left(c_v, \frac{V_{i,t}^{v,\text{src}}}{H_{i,t}^v} \right),$$

where:

$$c_B = 1.5, \quad c_{D^f} = c_{D^m} = 3.$$

Targets are excluded when:

- source counts are nonpositive or NaN;
- the denominator is below 1;
- population is below 10,000, when the population raster exists.

4.2 Migration targets

For eligible years from 1950 onward:

$$G_{i,t} = \frac{\Delta P_{i,t}}{h_t},$$
$$M_{i,t}^{\text{res}} = G_{i,t} - \left(B_{i,t}^{\text{UNWPP}} - D_{i,t}^{f,\text{UNWPP}} - D_{i,t}^{m,\text{UNWPP}} \right).$$

The migration target is:

$$y_{i,t}^M = \text{clip} \left(\frac{M_{i,t}^{\text{res}}}{P_{i,t}}, -1, 1 \right).$$

Pixels below population 1 are excluded. Missing event values are treated as zero.

Interval caveat: (h_t) is calculated from the method's current `years` list, with the first entry assigned 1. It is not read from the delta raster's metadata. Subsetting years can therefore change the inferred annual flow for the same input delta.

5. Shared regression and range calibration

This machinery is used for all four quantities:

$$v \in \{B, D^f, D^m, M\}.$$

5.1 Fit yearly regressions

The intended ridge-regression formulation is:

$$\hat{\beta}_t^v = \underset{\beta}{\operatorname{argmin}} \left[\sum_{i \in \mathcal{T}_t} \left(y_{i,t}^v - X_{i,\tau(t)}^\top \beta \right)^2 + \lambda \|\beta\|_2^2 \right].$$

The default (λ) is 1, in priors scaled by maximum order of magnitude of unit of measurement (condition number).

For a prediction year:

$$\eta_{i,t}^v = X_{i,\tau(t)}^\top \hat{\beta}_{\text{selected}}^v,$$

using the year-specific model if available, otherwise the unified model.

The unified model is produced by `geomeanOLSModels` with the callback:

$$w(b) = |b|.$$

Its exact coefficient-combination equation, including handling of negative coefficients, cannot be recovered without that helper.

5.2 Estimate target ranges

For each training year, take sampled target quantiles:

$$\ell_t^v = Q_{0.01}(y_t^v), \quad u_t^v = Q_{0.99}(y_t^v).$$

Across the years examined in that invocation:

$$\ell_*^v = \min_t \ell_t^v, \quad u_*^v = \max_t u_t^v.$$

Let (n^v_t) be the stored sample-size statistic and:

$$\gamma_t^v = \exp(-n_t^v/15).$$

The reconstruction bounds are:

$$L_t^v = (1 - \gamma_t^v) \ell_t^v + \gamma_t^v \ell_*^v,$$

$$U_t^v = (1 - \gamma_t^v) u_t^v + \gamma_t^v u_*^v.$$

For births and deaths, the lower bound is also clamped to zero.

Without local target statistics:

$$L_t^v = \ell_*^v, \quad U_t^v = u_*^v.$$

The sampling rules differ:

- births/deaths: every seventh pixel, retaining positive targets; stored sample size is seven times the sampled count;

- migration: every seventeenth pixel, retaining nonzero targets with population above 10,000; stored sample size is the sampled count.

5.3 Compress regression extremes

Across inhabited land pixels, calculate:

$$q_1 = Q_{0.25}(\eta), \quad q_3 = Q_{0.75}(\eta).$$

Set:

$$\alpha = \begin{cases} q_3 - q_1, & q_3 - q_1 > 10^{-5}, \\ \text{sd}(\eta), & \text{sd}(\eta) > 10^{-5}, \\ 0.01, & \text{otherwise.} \end{cases}$$

Define fences:

$$a = q_1 - 1.5\alpha, \quad b = q_3 + 1.5\alpha.$$

Apply:

$$R(x) = \begin{cases} a - \alpha \log \left(1 + \frac{a-x}{\alpha} \right), & x < a, \\ x, & a \leq x \leq b, \\ b + \alpha \log \left(1 + \frac{x-b}{\alpha} \right), & x > b. \end{cases}$$

Then min–max normalise:

$$z_{i,t}^v = \text{clip} \left(\frac{R(\eta_{i,t}^v) - R(\eta_{\min,t}^v)}{R(\eta_{\max,t}^v) - R(\eta_{\min,t}^v)}, 0, 1 \right).$$

If the denominator is zero, use (z=0.5).

Finally, reconstruct a multiplier or migration rate:

$$\theta_{i,t}^v = L_t^v + z_{i,t}^v (U_t^v - L_t^v).$$

Thus, the OLS prediction is not used directly as the final rate. It is transformed into a relative spatial score and remapped onto an empirical range.

6. Initial net migration and sex allocation

6.1 Convert the predicted rate to a count

$$M_{i,t}^{\text{pred}} = P_{i,t} \theta_{i,t}^M.$$

Where the UNWPP event files and delta file exist, replace this with the uncapped accounting residual:

$$M_{i,t}^{(0)} = \begin{cases} M_{i,t}^{\text{res}}, & \text{source files available,} \\ M_{i,t}^{\text{pred}}, & \text{otherwise.} \end{cases}$$

6.2 Balance global migration

Over pixels with positive population, calculate:

$$b_t = - \frac{\sum_{i:P_{i,t}>0} M_{i,t}^{(0)}}{\sum_{i:P_{i,t}>0} P_{i,t}}.$$

Then:

$$M_{i,t}^{(1)} = M_{i,t}^{(0)} + P_{i,t} b_t \quad (P_{i,t} > 0).$$

This gives, in ideal arithmetic over that mask:

$$\sum_{i:P_{i,t}>0} M_{i,t}^{(1)} = 0.$$

It also changes source-derived residuals, so those residuals no longer necessarily balance the unchanged source births and deaths locally.

6.3 Intended Rogers–Castro sex split

The schedule is:

$$R_s(x) = 0.02e^{-0.1x} + 0.06 \exp \left[-0.1(x - \mu_s) - e^{-0.3(x-\mu_s)} \right] + 0.003,$$

with:

$$\mu_f = 20, \quad \mu_m = 22.5.$$

These are location parameters, not the exact peak ages.

The intended potentials are:

$$Q_{i,t}^s = \sum_a N_{i,t,a}^s R_s(x_a).$$

Then:

$$\rho_{i,t}^f = \frac{Q_{i,t}^f}{Q_{i,t}^f + Q_{i,t}^m}, \quad \rho_{i,t}^m = 1 - \rho_{i,t}^f,$$

and:

$$M_{i,t}^{s,(1)} = \rho_{i,t}^s M_{i,t}^{(1)}.$$

7. Initial births and deaths

7.1 Reconstruct event counts

Before reconciliation:

$$B_{i,t}^{(0)} = \min(H_{i,t}^B \theta_{i,t}^B, 1.5 H_{i,t}^B),$$

$$D_{i,t}^{s,(0)} = \min(H_{i,t}^{D^s} \theta_{i,t}^{D^s}, P_{i,t}).$$

Implementation exception: if the normalised score ($z_{i,t}^v \leq 0$), the code leaves the count at zero. It does not reconstruct ($L_t^v H_{i,t}^v$), even when the lower bound is positive.

7.2 Redistribute total deaths between sexes

First retain total predicted deaths:

$$D_{i,t}^{(0)} = D_{i,t}^{m,(0)} + D_{i,t}^{f,(0)}.$$

Define a sex-allocation score from the independently normalised death predictions:

$$\pi_{i,t}^m = \begin{cases} \frac{z_{i,t}^{D^m}}{z_{i,t}^{D^m} + z_{i,t}^{D^f}}, & z_{i,t}^{D^m} + z_{i,t}^{D^f} > 0, \\ 0.5, & \text{otherwise.} \end{cases}$$

Then:

$$D_{i,t}^{m,(1)} = \pi_{i,t}^m D_{i,t}^{(0)},$$

$$D_{i,t}^{f,(1)} = (1 - \pi_{i,t}^m) D_{i,t}^{(0)}.$$

This is a heuristic allocation rule. The normalised scores are not independently established male/female death probabilities.

8. Reconcile births, deaths, and migration

This stage runs only when the required migration files, population delta, and all three working event arrays are available.

Suppress pixel and year subscripts below.

8.1 Calculate the natural-increase discrepancy

Required natural increase, conditional on migration:

$$N_{\text{required}} = G - M^{(1)}.$$

Predicted natural increase:

$$N_{\text{predicted}} = B^{(0)} - D^{(0)}.$$

Discrepancy:

$$e = G - M^{(1)} - (B^{(0)} - D^{(0)}).$$

8.2 Increase births or deaths

The code never reduces an event count to absorb this discrepancy. Instead:

$$B^* = B^{(0)} + \max(e, 0),$$

$$D^* = D^{(0)} + \max(-e, 0).$$

Before caps, these satisfy:

$$B^* - D^* + M^{(1)} = G.$$

Interpretation:

- positive discrepancy -> increase births;
- negative discrepancy -> increase deaths.

This is an allocation assumption, not evidence that the discrepancy actually represents a baby boom or mortality crisis.

8.3 Apply coded caps

$$B^{(2)} = \text{clip}(B^*, 0, 1.5H^B),$$

$$D^{(2)} = \text{clip}(D^*, 0, P).$$

These are modelling caps, not universal biological identities.

Reallocate deaths using the pre-reconciliation death split:

$$D^{m,(2)} = D^{(2)} \frac{D^{m,(1)}}{D^{(0)}},$$

$$D^{f,(2)} = D^{(2)} \frac{D^{f,(1)}}{D^{(0)}},$$

when $(D^{(0)} > 0)$. Otherwise the split is 0.512 male and 0.488 female.

8.4 Return the remaining discrepancy to migration

The migration required after capping is:

$$M^* = G - (B^{(2)} - D^{(2)}).$$

Define:

$$\delta M = M^* - M^{(1)}.$$

The actual update is thresholded:

$$M^{(2)} = \begin{cases} M^*, & |\delta M| > 1, \\ M^{(1)}, & |\delta M| \leq 1. \end{cases}$$

When updated, sex-specific migration receives the same proportional adjustment:

$$M^{s,(2)} = M^{s,(1)} + \rho^s \delta M,$$

where the ratio is taken from the existing sex-specific migration totals, or falls back to 0.488/0.512.

There is no second global zero-sum balancing step.

9. Final source override

After reconciliation, births and deaths are written using:

$$V_{i,t}^{v,\text{final}} = \begin{cases} 0, & P_{i,t} \leq 0, \\ V_{i,t}^{v,\text{src}}, & V_{i,t}^{v,\text{src}} > 0 \text{ and not NaN}, \\ V_{i,t}^{v,(2)}, & \text{otherwise,} \end{cases}$$

for:

$$v \in \{B, D^f, D^m\}.$$

Migration is not recomputed after these source overrides.

Therefore, even if the working arrays satisfied:

$$G = B^{(2)} - D^{f,(2)} - D^{m,(2)} + M^{(2)},$$

the saved source-overridden outputs may not.

The system in compact form

Ignoring missing-data branches and implementation exceptions, the intended reconstruction is:

$$\begin{aligned} N_a^s &= P \frac{\text{SexPrior}_s [\text{PAVA} (\text{MNL}(X))]_a}{\text{sum across sexes and ages}}, \\ H^B &= N_{00}^f + N_{00}^m, \\ H^{D^s} &= \sum_a N_a^s \mu_a \left[\text{PAVA} \left(\frac{N_a^s}{w_a} e^{g x_a} \right) \right], \\ B^{(0)} &= H^B \theta^B(X), \\ D^{s,(0)} &= H^{D^s} \theta^{D^s}(X), \\ M^{(0)} &= P \theta^M(X), \\ M^{(1)} &= \text{GlobalBalance}(M^{(0)}), \\ e &= G - M^{(1)} - (B^{(0)} - D^{(0)}), \\ B^{(2)} &= \text{clip} \left(B^{(0)} + \max(e, 0), 0, 1.5 H^B \right), \\ D^{(2)} &= \text{clip} \left(D^{(0)} + \max(-e, 0), 0, P \right), \\ M^{(2)} &\approx G - B^{(2)} + D^{(2)}. \end{aligned}$$

Source substitutions occur at several points, including after the last reconciliation.

Population totals determine scale; regression estimates spatial variation; age-shape and sex-ratio priors determine structure; infant stock and age-profile decline provide event-count denominators; and the demographic accounting residual allocates remaining discrepancies.

The current code does not guarantee absolute compliance in final rasters.